suitable rotation (dictated, as you can see, by which result I find), to put Photon 3 into the desired state.

4. In fact, the mathematically inclined reader will note that it is not hard to prove the so-called no-quantum-cloning theorem. Imagine we have a unitary cloning operator U that takes any given state as input and produces two copies of it as output (U maps $|\alpha\rangle \rightarrow |\alpha\rangle|\alpha\rangle$, for any input state $|\alpha\rangle$). Note that U acting on a state like $(|\alpha\rangle + |\beta\rangle)$ yields $(|\alpha\rangle|\alpha\rangle + |\beta\rangle|\beta\rangle)$, which is not a two-fold copy of the original state $(|\alpha\rangle + |\beta\rangle)(|\alpha\rangle + |\beta\rangle)$, and hence no such operator U exists to carry out quantum cloning. (This was first shown by Wootters and Zurek in the early 1980s.)

5. Many researchers have been involved in developing both the theory and the experimental realization of quantum teleportation. In addition to those discussed in the text, the work of Sandu Popescu while at Cambridge University played an important part in the Rome experiments, and Jeffrey Kimble's group at the California Institute of Technology has pioneered the teleportation of continuous features of a quantum state, to name a few.

6. For extremely interesting progress on entangling many-particle systems, see, for example, B. Julsgaard, A. Kozhekin, and E. S. Polzik, "Experimental long-lived entanglement of two macroscopic objects," *Nature* 413 (Sept. 2001), 400–403.

7. One of the most exciting and active areas of research making use of quantum entanglement and quantum teleportation is the field of quantum computing. For recent general-level presentations of quantum computing, see Tom Siegfried, *The Bit and the Pendulum* (New York: John Wiley, 2000), and George Johnson, *A Shortcut Through Time* (New York: Knopf, 2003).

8. One aspect of the slowing of time at increasing velocity, which we did not discuss in Chapter 3 but will play a role in this chapter, is the so-called twin paradox. The issue is simple to state: if you and I are moving relative to one another at constant velocity, I will think your clock is running slow relative to mine. But since you are as justified as I in claiming to be at rest, you will think that mine is the moving clock and